Indian Equities Analytics & Direction Predictor

Capstone Final Report

Universe: RELIANCE.NS | TCS.NS | HDFCBANK.NS | INFY.NS | ICICIBANK.NS

Reviewer: Ravi Agarwal

1. Problem Statement

This capstone frames "will this stock go up or down tomorrow?" as a supervised binary classification problem for 5 Indian large-cap stocks, using only historical price/volume data, and builds a complete reproducible pipeline around it: data ingestion, feature engineering, model training/selection, and an interactive dashboard.

The goal is not a production trading signal, but a professionally-structured ML workflow end to end, with an honest account of where such a model succeeds and where it fundamentally struggles.

2. Data

Ticker	Company	Sector
RELIANCE.NS	Reliance Industries	Energy / Conglomerate
TCS.NS	Tata Consultancy Services	IT
HDFCBANK.NS	HDFC Bank	Banking
INFY.NS	Infosys	IT
ICICIBANK.NS	ICICI Bank	Banking

Source: Yahoo Finance (yfinance), 5 years of daily OHLCV data per ticker, cached to data/raw/. If yfinance/internet is unavailable, src/data.py deterministically falls back to a synthetic OHLCV generator so the pipeline remains fully reproducible offline, swappable for real data with zero code changes.

3. Methodology

3.1 Exploratory Data Analysis

We examined normalized price performance, return distributions and volatility, cross-stock return correlation, and each engineered feature's distribution split by the next-day-direction target. Class balance was confirmed close to 50/50 for every stock, as expected for liquid large-caps.

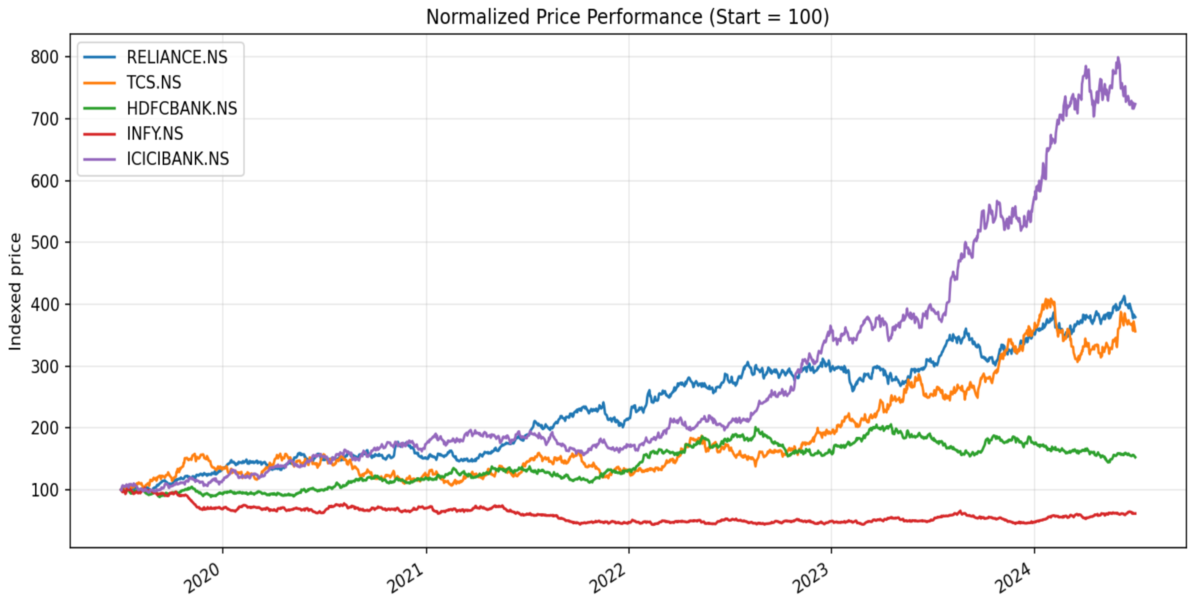

Figure 1. Normalized price performance, all 5 stocks (start = 100).

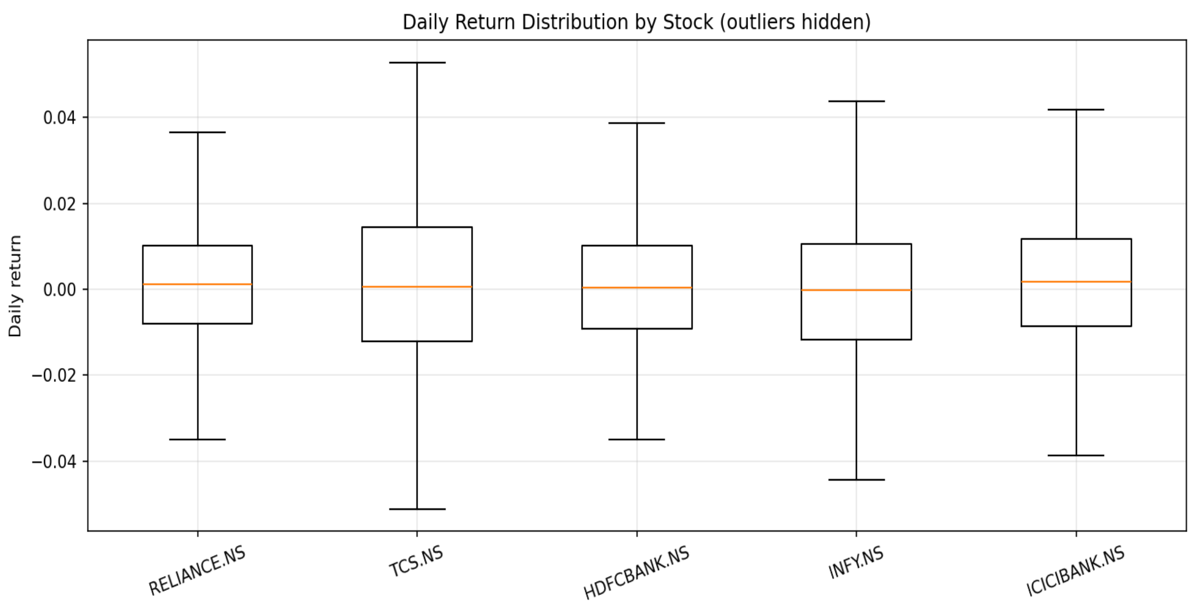

Figure 2. Daily return distribution by stock.

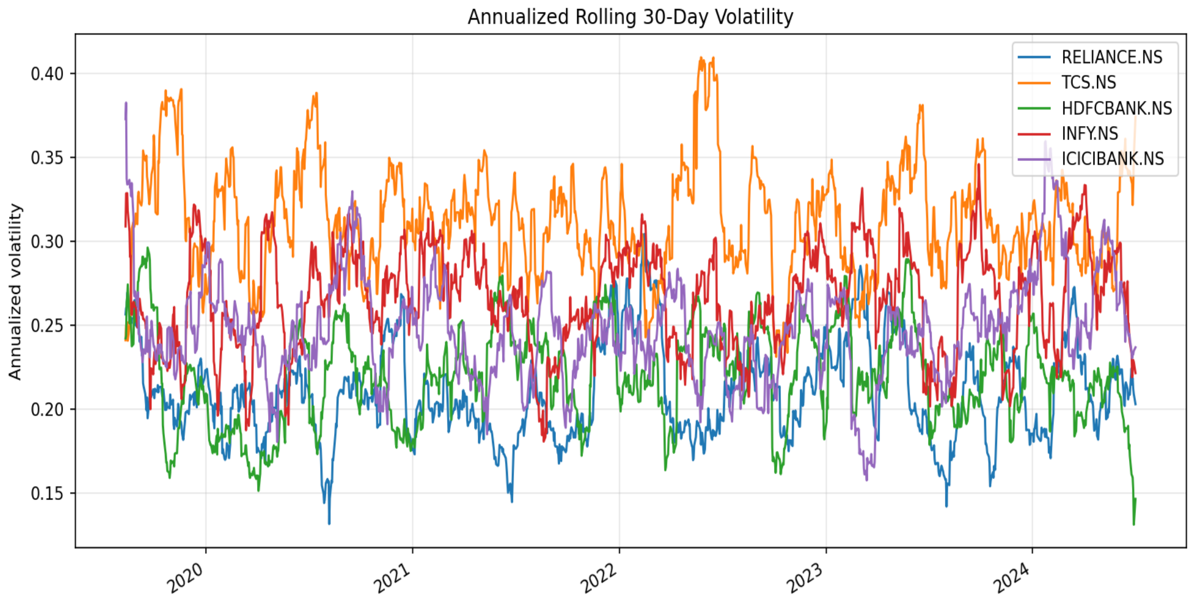

Figure 3. Rolling 30-day annualized volatility.

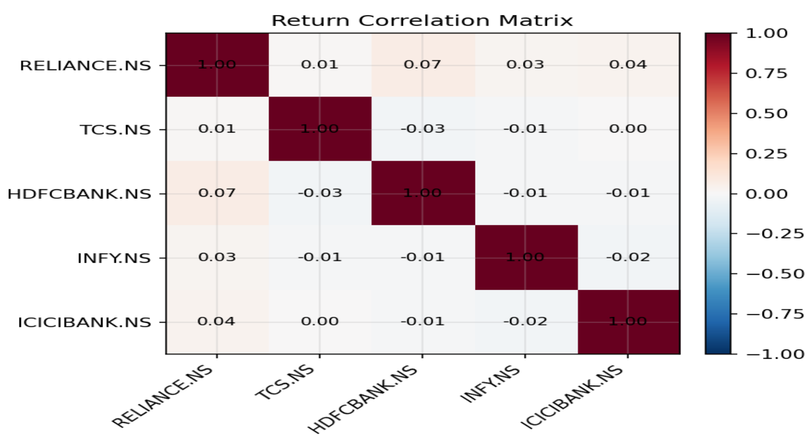

Figure 4. Return correlation matrix across the 5 stocks.

3.2 Feature Engineering

Ten technical-indicator features were engineered per stock from OHLCV data only (no look-ahead leakage): 1-day return and log-return, price-to-SMA(10) and price-to-SMA(50) ratios, price-to-EMA(20) ratio, 14-day RSI, MACD histogram, normalized Bollinger Band width, rolling 10-day return volatility, and 1-day volume change. Target = 1 if next day's close is higher than today's, else 0.

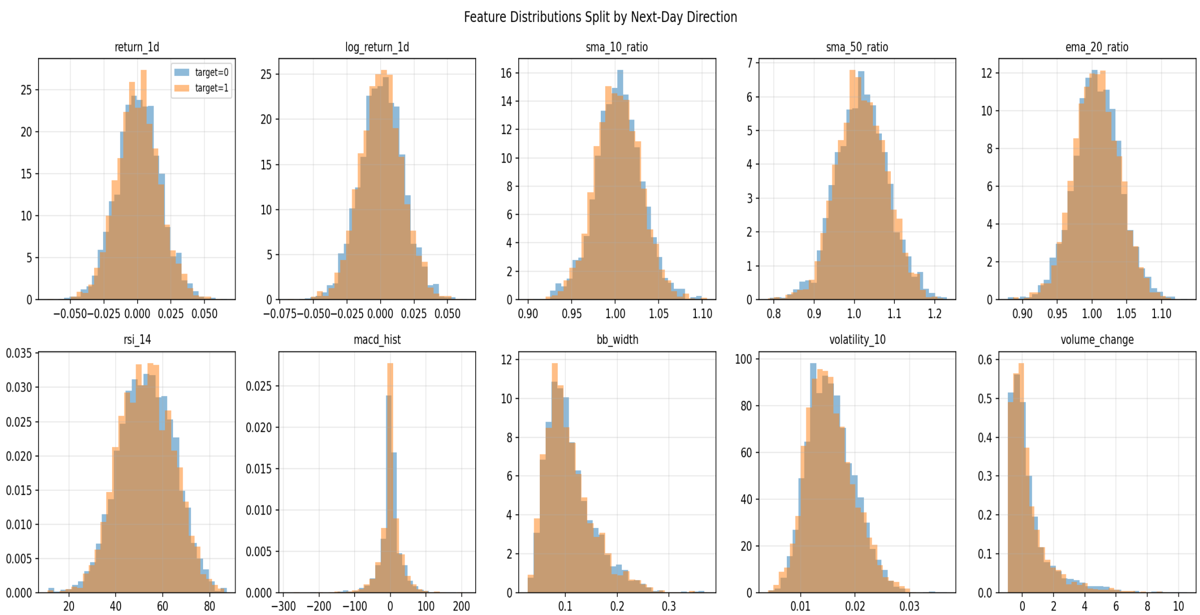

Figure 5. Feature distributions split by next-day direction.

3.3 Modeling

Three classifiers were trained and compared: Logistic Regression (scaled, balanced classes) as a linear baseline; Random Forest (300 depth-limited trees, balanced classes); and Gradient Boosting (200 shallow trees). All were evaluated with 5-fold TimeSeriesSplit cross-validation — chronological folds only, so each fold trains strictly on the past and tests on unseen future data, avoiding look-ahead bias. The highest-mean-CV-accuracy model was refit on the full dataset and persisted with joblib.

3.4 Dashboard

An interactive Streamlit app (app.py) lets a user pick any of the 5 stocks and see a candlestick price chart with SMA/EMA overlays, RSI/MACD/Bollinger indicator panels, the model's next-day direction prediction with probability, and a table of historical cross-validated accuracy for all 3 models.

4. Results

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.507	0.524	0.539	0.531
Random Forest	0.501	0.517	0.531	0.523
Gradient Boosting	0.507	0.520	0.610	0.561

Best model selected: Gradient Boosting (highest mean cross-validated accuracy; ties broken toward higher F1). Exact figures regenerate on every `python run_pipeline.py` run — see models/metrics.json for current values.

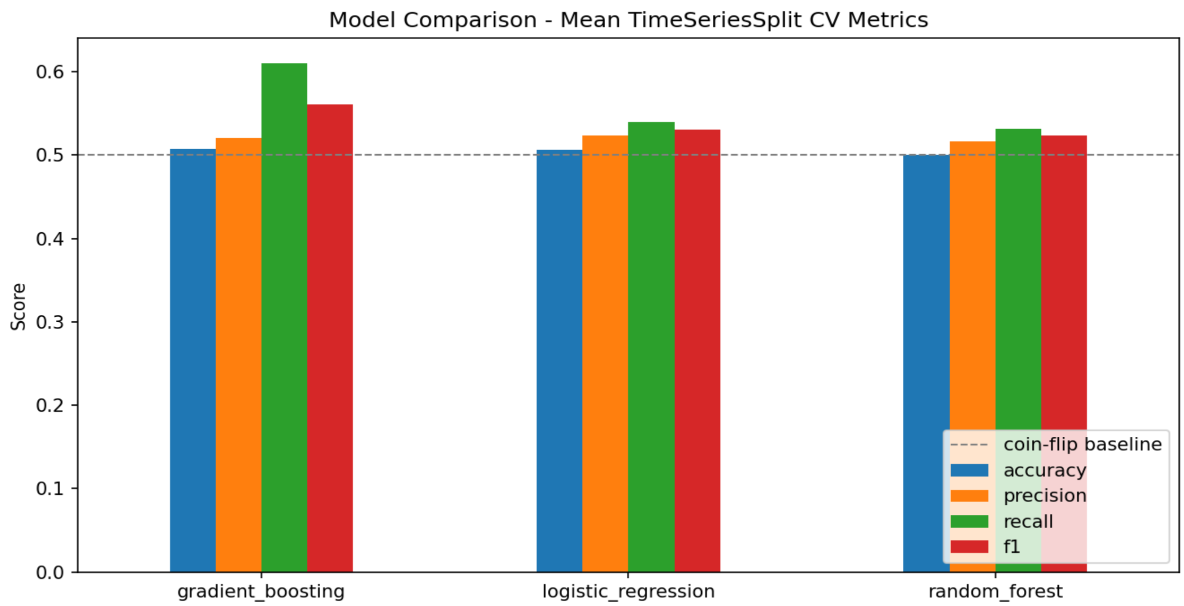

Figure 6. Model comparison across CV metrics.

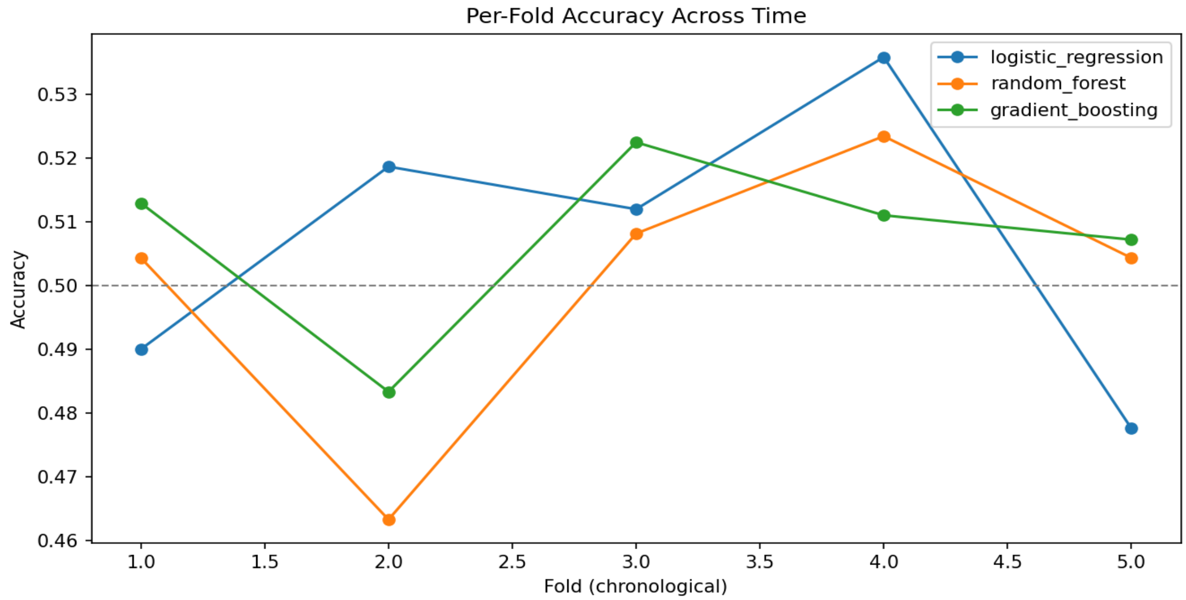

Figure 7. Per-fold accuracy across chronological folds.

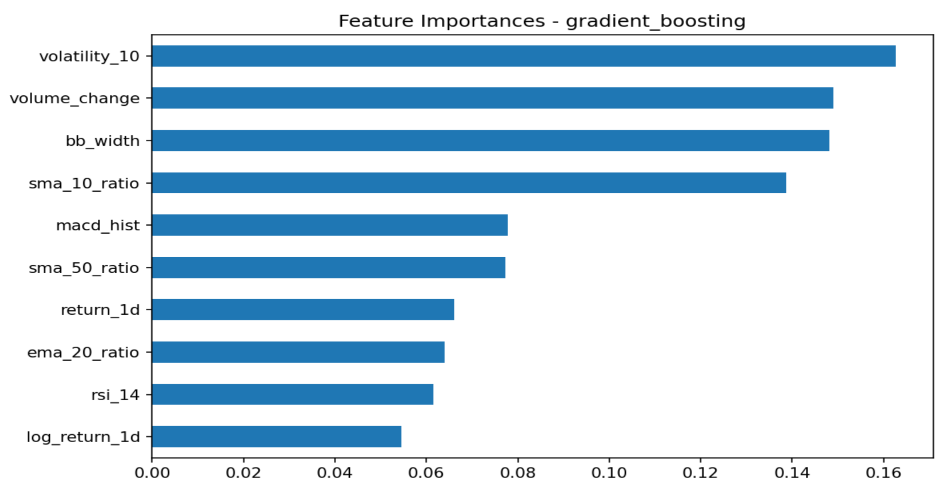

Figure 8. Feature importances for the selected model.

5. Limitations

• **Accuracy near the random-walk baseline.** Short-horizon (1-day) direction prediction on liquid, well-covered large-caps is close to weak-form market-efficient — historical price/volume alone carries limited signal for tomorrow's direction. This matches published findings, not a pipeline defect.

• **No fundamental, macro, or sentiment data** — only technical indicators from price/volume were used.

• **Binary framing loses information** about move magnitude and confidence (though predicted probability is exposed in the dashboard).

• **Per-stock, not portfolio-level** — no cross-stock, sector, or market-relative features.

• **No transaction-cost-aware backtest** — accuracy alone does not imply profitability.

6. Future Work

• Add fundamental data (EPS, P/E), macro variables (Nifty 50, USD/INR, crude oil for RELIANCE), and news/sentiment features.

• Predict a probability/confidence score and size a signal accordingly, rather than a hard binary label.

• Extend to multi-day or weekly horizons.

• Add a walk-forward backtest with transaction costs and realistic position sizing.

• Explore sequence models (LSTM/Transformer) as a comparison point.

• Add a rolling monthly retraining schedule to simulate production deployment.

7. Conclusion

This capstone delivers a fully reproducible, single-command pipeline covering data ingestion, feature engineering, sound time-series-aware model comparison, and an interactive dashboard. The headline result — that next-day direction on large-cap Indian equities is only marginally more predictable than a coin flip using technical indicators alone — is itself a meaningful, honestly-reported finding, and the codebase is structured to make extending toward more predictive feature sets straightforward.